

Quadrature Down Converter

Analog Electronic Circuits (EC2.103) — Spring 2026, IIIT Hyderabad

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Abstract—This paper presents the complete design, analysis, and LTspice simulation of a Quadrature Down Converter (QDC) for the Analog Electronic Circuits course project (Spring 2026). The system comprises three tightly integrated subsystems: a UA741-based quadrature oscillator with 1N4148 diode amplitude stabilization operating at 150 kHz, dual NMOS switching mixers (TSMC 180 nm, $W = 10 \mu\text{m}$), and RC low-pass filters with a -3dB cutoff of $\approx 3.39\text{ kHz}$ followed by an output amplifier stage. The oscillator generates two quadrature sinusoidal LO signals (v_{OSCI} and v_{OSCQ}) with a 90° phase difference. Each NMOS mixer multiplies the 100 mV RF input with the respective LO signal, producing IF components. After low-pass filtering, the final I and Q outputs at $f_{IF} = |f_{osc} - f_{in}|$ maintain a 90° phase difference, confirming successful quadrature downconversion.

Index Terms—quadrature down converter, QDC, NMOS mixer, quadrature oscillator, UA741, TSMC 180nm, downconversion, low-pass filter, IQ demodulation

I. INTRODUCTION AND NEED FOR QUADRATURE DOWNCONVERSION (Q1)

A. Motivation

In modern wireless receivers (Bluetooth, Wi-Fi, WLAN), a high-frequency RF signal $v_{in} = A_1 \cos(\omega_{in}t)$ must be translated to a lower intermediate frequency (IF) for practical baseband processing. Simple real-valued downconversion by multiplying with a single LO produces:

$$v_{in} \cdot \cos(\omega_{osc}t) = \frac{A_1}{2} [\cos((\omega_{in} - \omega_{osc})t) + \cos((\omega_{in} + \omega_{osc})t)] \quad (1)$$

Both the desired signal at $\omega_{in} - \omega_{osc}$ and its image $\omega_{osc} - \omega_{in}$ are indistinguishable, causing image interference. Furthermore, for asymmetrically modulated signals downconverted to zero-IF, the signal can interfere with itself.

B. Quadrature Solution

A QDC solves this by generating two orthogonal LO cop $v_{OSCI} = A_2 \cos(\omega_{osc}t)$ and $v_{OSCQ} = A_2 \sin(\omega_{osc}t)$. Mix produces:

$$v_{IFI} = v_{in} \cdot v_{OSCI} = \frac{A_1 A_2}{2} [\cos(\omega_{in}t - \omega_{osc}t) + \cos(\omega_{in}t + \omega_{osc}t)] \quad (2)$$

$$v_{IFQ} = v_{in} \cdot v_{OSCQ} = \frac{A_1 A_2}{2} [\sin(\omega_{in}t + \omega_{osc}t) - \sin(\omega_{in}t - \omega_{osc}t)] \quad (3)$$

After low-pass filtering, the IF components at $\omega_{IF} = \omega_{IN} - \omega_{OSC}$ are retained. Because v_{IFI} contains $\cos(\omega_{IF}t)$ and v_{IFQ} contains $-\sin(\omega_{IF}t)$, the two outputs are inherently 90° out of phase. The complex baseband signal $v_{IFI} + jv_{IFQ}$ fully characterizes the original RF signal's amplitude and phase, enabling image rejection and complete demodulation [?] [?].

For example, with $f_{IN} = 2.4\text{ GHz}$ and $f_{OSC} = 2.401\text{ GHz}$, one obtains $f_{IF} = 1\text{ MHz}$ — a manageable baseband frequency.

II. SYSTEM ARCHITECTURE

The complete QDC block diagram is shown in Fig. 1. The RF input enters two parallel switch-mixer branches. A shared quadrature oscillator feeds v_{OSCI} (cosine) to the I-branch mixer and v_{OSCQ} (sine) to the Q-branch mixer. Each mixer output passes through a low-pass filter and amplifier to yield the final I_{FINAL} and Q_{FINAL} outputs. The LTspice simulation uses the following directives:

```
.tran 0 1m 0 10u
.inc UA741.301
.inc TSMC_180nm.txt
.param width=10u
```

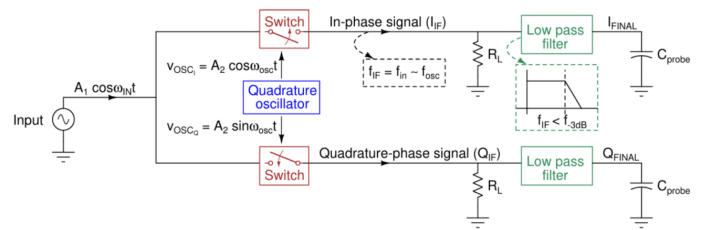


Figure 1: Block Diagram of QDC

Fig. 1. Block diagram of the Quadrature Down Converter.

III. QUADRATURE OSCILLATOR DESIGN (Q2)

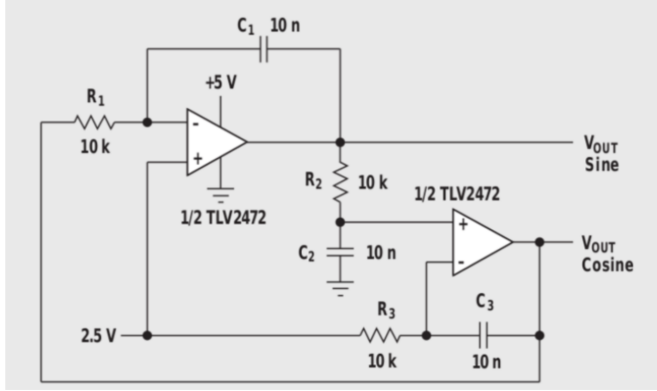


Fig. 2. Schematic of the quadrature oscillator.

A. Topology

The quadrature oscillator uses two UA741 op-amps (U1, U2) in a two-integrator feedback loop. Six 1N4148 diodes (D1–D6) arranged in back-to-back pairs provide soft amplitude limiting, ensuring a stable $1V_{PP}$ output without saturation distortion. The supply voltages are $V_{DD} = +12V$ and $V_{SS} = -12V$ for the UA741 op-amps, consistent with their rated supply range and the UA741.301 model file.

B. Component Value Calculations

Targeting $f_{osc} = 150\text{ kHz}$ and choosing $C = 182\text{ pF}$:

$$R = \frac{1}{2\pi f_{osc} C} = \frac{1}{2\pi \times 150 \times 10^3 \times 182 \times 10^{-12}} \approx 5.83\text{ k}\Omega \quad (4)$$

The practical value chosen is $R_{10} = 10\text{ k}\Omega$. Due to the finite gain-bandwidth of the UA741 ($GBW \approx 1\text{ MHz}$), the effective oscillation frequency deviates from theory, so the feedback resistor is trimmed to $R_9 = 326\Omega$ to achieve the target amplitude and frequency.

The diode clipping network limits the output to approximately $\pm 0.5V$ (one diode drop), keeping the oscillation amplitude near $1V_{PP}$.

C. Component Values

TABLE I
QUADRATURE OSCILLATOR COMPONENT VALUES

Component	Value
Op-amps U1, U2	UA741
Supply V_{DD} / V_{SS}	+12 V / -12 V
R_9 (amplitude trim)	326 Ω
R_{10} (frequency-setting)	10 k Ω
C_2	182 pF
C_3	200 pF
C_4, C_5	1 nF each
Diodes D1–D6	1N4148

D. Working Principle and Barkhausen Criterion

For the two-integrator loop, the loop gain with $R_i C_i = RC$ is:

$$A\beta = -\frac{1}{j\omega RC} \cdot \frac{1}{1 + j\omega RC} \cdot \left(1 + \frac{1}{j\omega RC}\right) \quad (5)$$

At $\omega = 1/RC$, $|A\beta| = 1$ and the total phase shift is 0° , satisfying the Barkhausen criterion for sustained oscillation. The two output nodes exhibit a 90° phase difference because the feedback path splits into a low-pass branch $H_{LP} = \frac{1}{1 + j\omega RC}$ and a high-pass branch $H_{HP} = \frac{j\omega RC}{1 + j\omega RC}$:

$$\angle H_{LP}|_{\omega=1/RC} = -45^\circ \quad (6)$$

$$\angle H_{HP}|_{\omega=1/RC} = +45^\circ \quad (7)$$

giving a phase difference of 90° between v_{OSCI} and v_{OSCQ} .

E. LTspice Schematic and Simulation Results

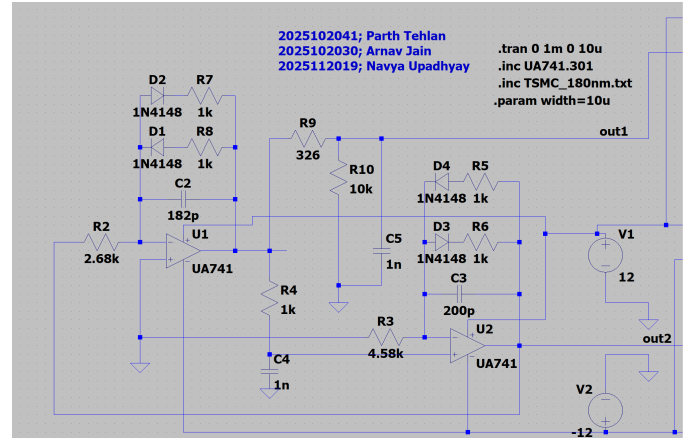


Fig. 3. LTspice schematic of the quadrature oscillator.

Inference: The oscillator produces two sinusoidal signals of $\approx 1V_{PP}$ amplitude at the target frequency of 145 kHz with a phase difference of approximately 90° , as required.

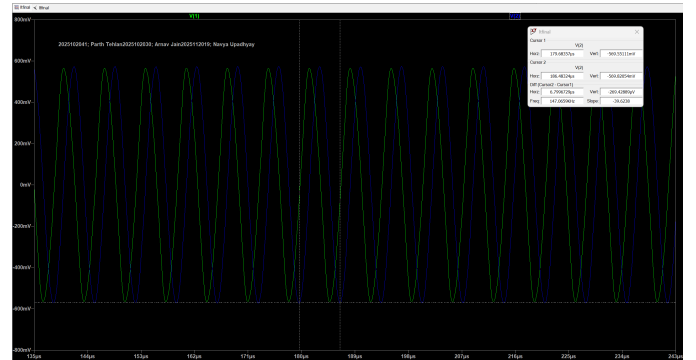


Fig. 4. Transient Response of Oscillator

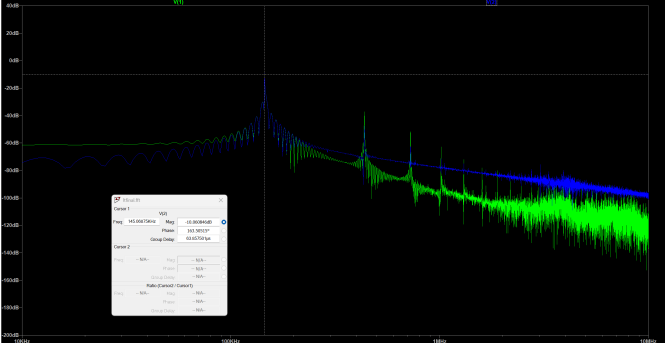


Fig. 5. FFT of v_{OSCI} and v_{OSCQ} confirming oscillation at $f_{osc} \approx 145$ kHz.

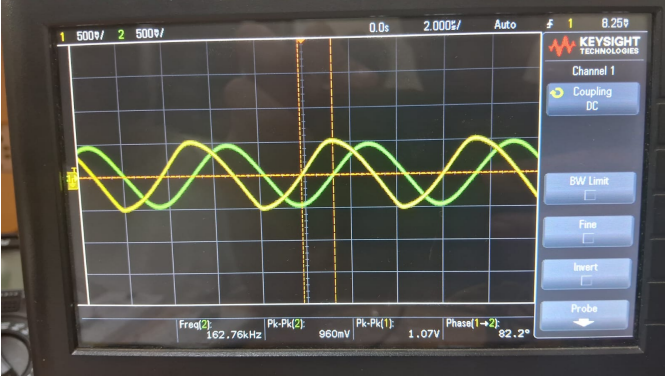


Fig. 6. Oscillator Output

IV. MIXER (SWITCH) DESIGN (Q3)

A. Circuit Description and MOSFET Parameters

Each mixer uses an NMOS transistor (M7 for I-channel, M8 for Q-channel) from the TSMC 180 nm process as a switching element. The oscillator signal is AC-coupled to the gate via C_c , while R_{bias} sets the DC gate voltage. The RF input v_{in} is applied at the source, and the IF output v_{IF} is taken at the drain.

The MOSFET parameters used in LTspice are:

TABLE II
NMOS MIXER TRANSISTOR PARAMETERS (TSMC 180 NM)

Parameter	Symbol	Value
Channel width	W	$10 \mu\text{m}$
Channel length	L	180 nm
Source area	A_S	$W \times 0.5 \mu\text{m} = 5 \mu\text{m}^2$
Drain area	A_D	$W \times 0.5 \mu\text{m} = 5 \mu\text{m}^2$
Source perimeter	P_S	$2W + 1 \mu\text{m} = 21 \mu\text{m}$
Drain perimeter	P_D	$2W + 1 \mu\text{m} = 21 \mu\text{m}$
Bias voltage	V_{bias}	1 V
Bias resistor	R_{bias}	$100 \text{ k}\Omega$
Coupling capacitor	C_c	$1 \mu\text{F}$
Load resistor	R_L	$1 \text{ k}\Omega$

The LTspice MOSFET instance line is:

M7 drain gate source 0 CMOSN

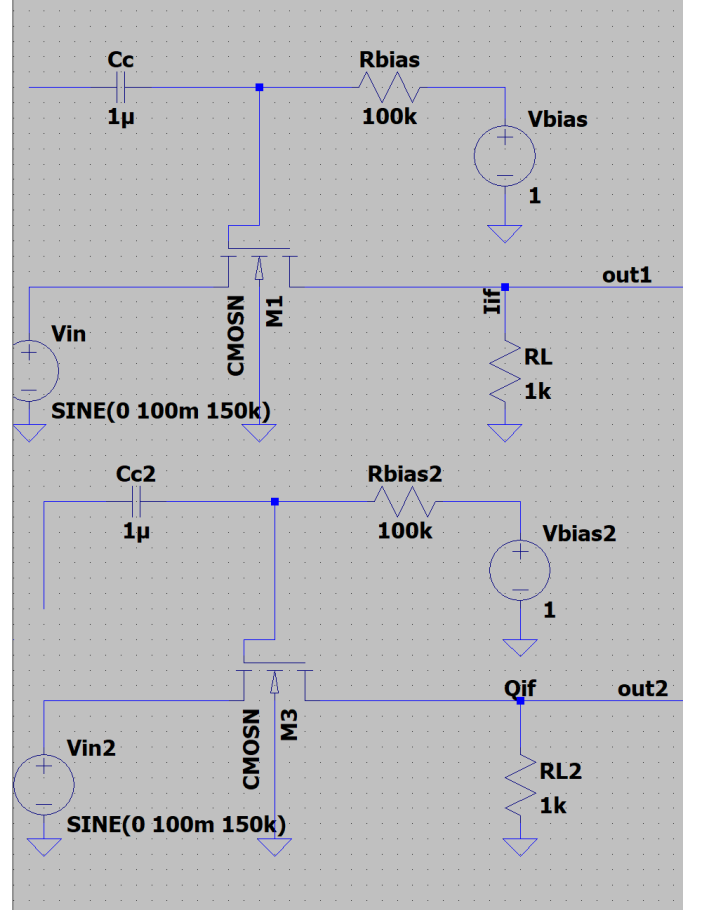


Fig. 7. LT Spice Schematic of Mixer

+ W=10u L=180n
+ AS=2.7e-12 AD=2.7e-12
+ PS=10.54e-6 PD=10.54e-6

B. Working Principle

The NMOS operates as a switch controlled by v_{OSCI} (or v_{OSCQ}). When $V_{GS} = v_{osc} + V_{bias} - v_{out} > V_{TH}$, the device is on and the input passes; when $V_{GS} < V_{TH}$, it is off. In the triode region:

$$I_D = \frac{1}{2} K_n [2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2] \quad (8)$$

For small V_{DS} and $V_{bias} \approx V_{TH}$, the output reduces to:

$$V_{out}(t) \approx R_L K_n V_{osc}(t) V_{in}(t) \quad (9)$$

This is equivalent to multiplication of v_{in} with the switching function $s(t)$, whose Fourier series is $s(t) = \sum_{k=\text{odd}} \frac{1}{k} \sin(2\pi k f_{osc} t)$. The output spectrum contains components at $|k f_{osc} \pm f_{in}|$. After low-pass filtering, only $f_{IF} = |f_{osc} - f_{in}|$ survives.

C. Simulation Results for $f_{osc} = 150 \text{ kHz}$

The RF input is $v_{in} = 100 \text{ mV}$ at six test frequencies with $f_{osc} = 150 \text{ kHz}$:

TABLE III
MIXER TEST FREQUENCIES AND EXPECTED IF

f_{in} (kHz)	$f_{IF} = f_{osc} - f_{in} $ (kHz)
145	5
148	2
149	1
151	1
152	2
155	5

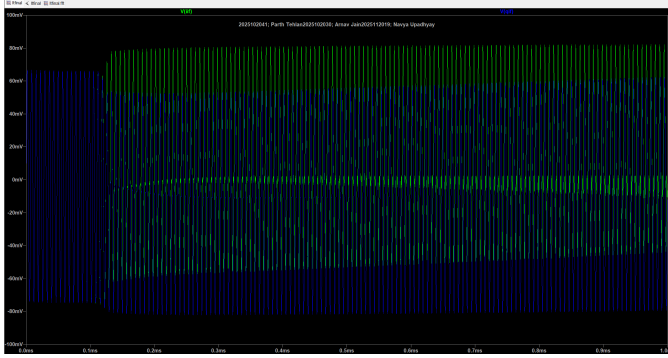


Fig. 8. Transient: v_{in} and v_{IF} for $f_{in} = 145$ kHz.

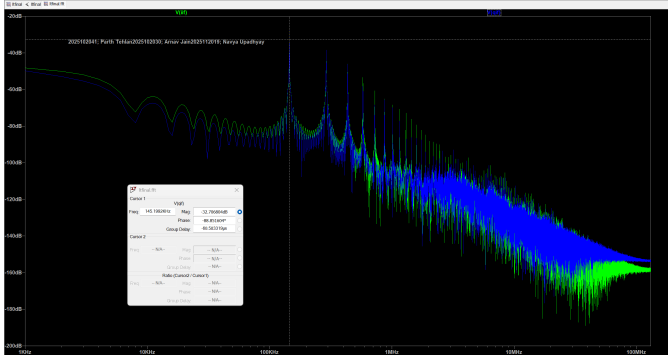


Fig. 9. FFT of v_{IF} for $f_{in} = 145$ kHz

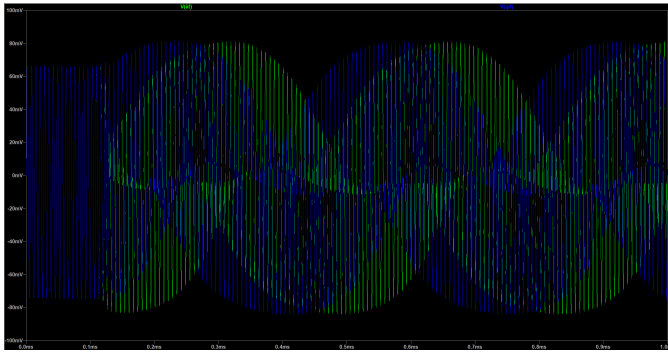


Fig. 10. Transient: v_{in} and v_{IF} for $f_{in} = 148$ kHz.

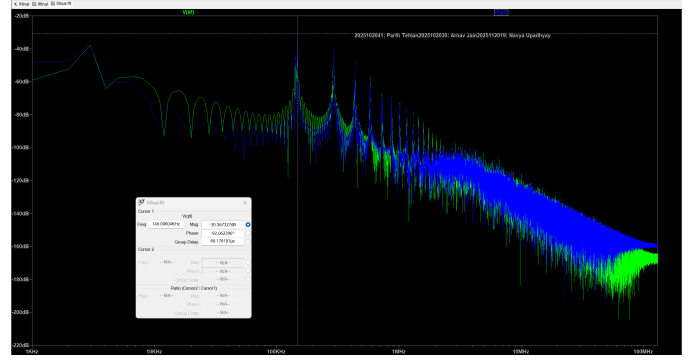


Fig. 11. FFT of v_{IF} for $f_{in} = 148$ kHz

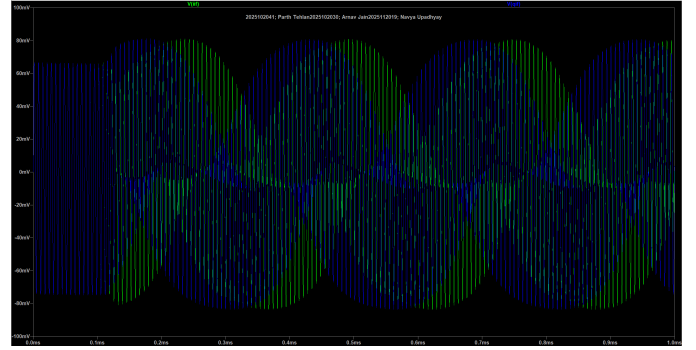


Fig. 12. Transient: v_{in} and v_{IF} for $f_{in} = 149$ kHz.

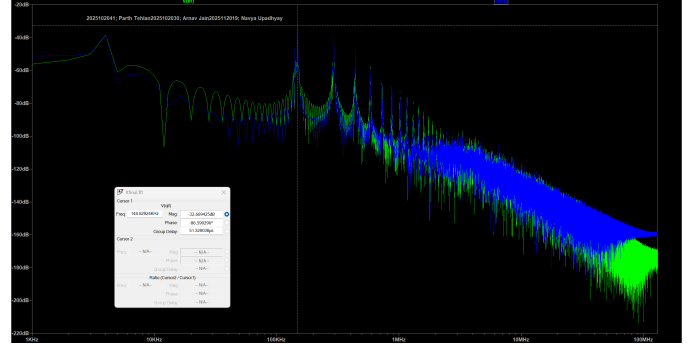


Fig. 13. FFT of v_{IF} for $f_{in} = 149$ kHz

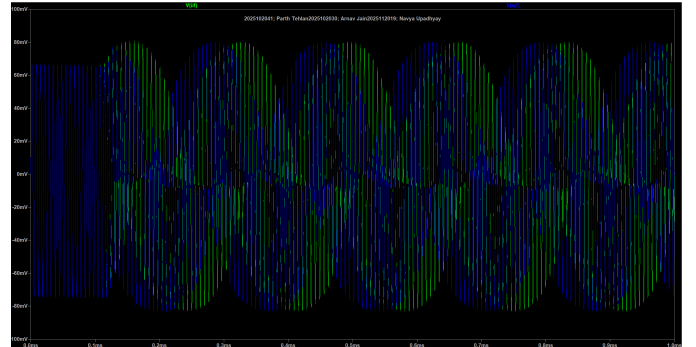


Fig. 14. Transient: v_{in} and v_{IF} for $f_{in} = 151$ kHz.

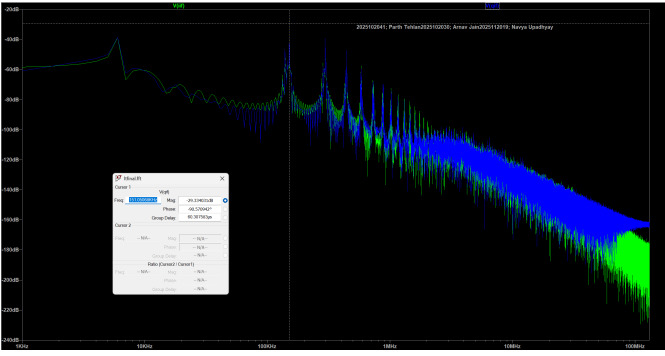


Fig. 15. FFT of v_{IF} for $f_{in} = 151$ kHz

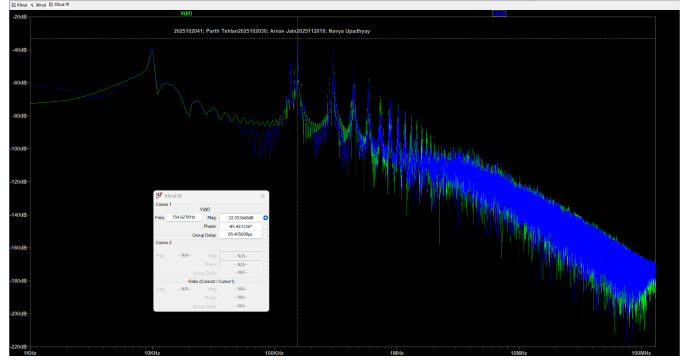


Fig. 19. FFT of v_{IF} for $f_{in} = 155$ kHz

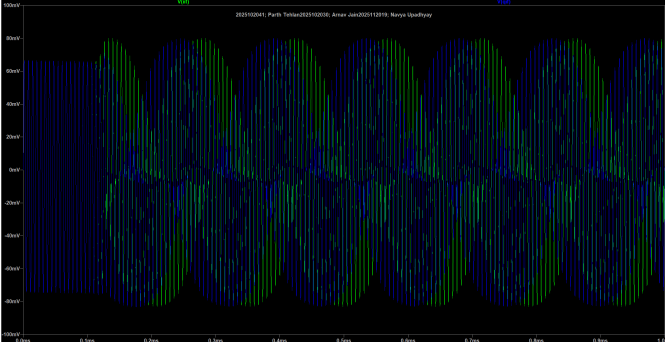


Fig. 16. Transient: v_{in} and v_{IF} for $f_{in} = 152$ kHz.

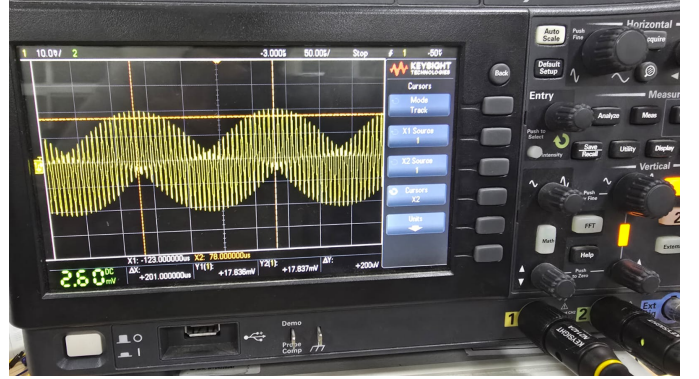


Fig. 20. Mixer Output

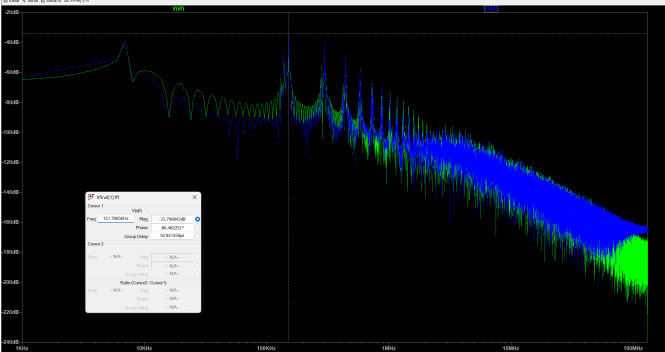


Fig. 17. FFT of v_{IF} for $f_{in} = 152$ kHz

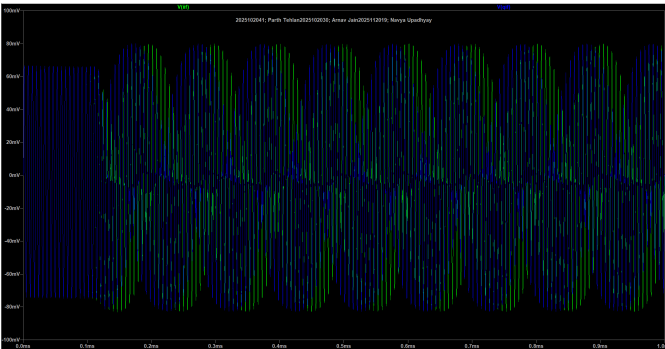


Fig. 18. Transient: v_{in} and v_{IF} for $f_{in} = 155$ kHz.

Circuit: The hardware realization of the NMOS mixer circuit was conducted using an available discrete NMOS transistor in the laboratory, noting that its physical characteristics and parasitic parameters naturally differ from the TSMC 180nm model utilized in the LTSpice simulations. Two distinct function generators were employed to supply the input signal (v_{in}) and the local oscillator signal (v_{osc}) to the source and gate terminals, respectively. The resulting intermediate frequency signal (v_{IF}) at the drain terminal was captured.

Inference: The FFT of v_{IF} shows a dominant peak at $|f_{osc} - f_{in}|$ for each test frequency, confirming correct switching-mixer operation. The IF amplitude decreases slightly for frequencies farther from f_{osc} (i.e., 145 kHz and 155 kHz) due to the roll-off of the output RC network.

V. LOW-PASS FILTER DESIGN (Q4)

A. Design and Component Calculation

A passive RC low-pass filter is designed using standard component values of $R = 470 \Omega$ and $C = 100$ nF. The -3 dB cutoff frequency f_c is calculated as:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 470 \times 100 \times 10^{-9}} \approx 3.39 \text{ kHz} \quad (10)$$

The transfer function for the circuit is:

$$H(j\omega) = \frac{1}{1 + j\omega RC}, \quad |H| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad (11)$$

At $f \approx 3.39$ kHz: $|H| = \frac{1}{\sqrt{2}} \approx -3$ dB.

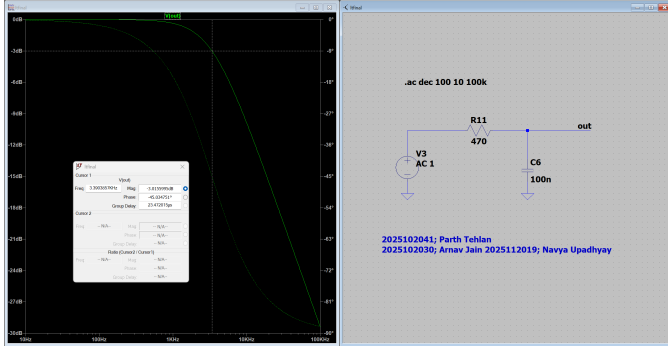


Fig. 21. LT Spice Schematic of Low Pass Filter



Fig. 22. Output of Circuit + LPF

B. Amplifier Stage ($Q4f$)

To obtain the specified output amplitude of ≈ 200 mV (400 mV_{PP}), a non-inverting amplifier stage is implemented as shown:

$$A_v \approx \frac{R_C}{R_E} = \frac{1 \text{ k}\Omega}{470 \Omega} \approx 2.1 \quad (12)$$

The additional capacitor network (C_{16} , $C_{17} = 1 \mu\text{F}$; C_{19} , $C_{20} = 22 \mu\text{F}$) provides output AC coupling and ensures the DC operating point does not appear at the final output.

While implementing, on trying the same circuit and changing resistance/capacitance values, this amplifier did not work. We also tried BJT Common Emitter stage, but due to non-idealities, it was not able to give desired gain.

C. Simulation Results

Inference: The LPF passes the 1 kHz signal with negligible attenuation and attenuates the 10 kHz signal significantly, confirming the 3 kHz cutoff design. When cascaded with the mixer, higher harmonics at $|kf_{osc} \pm f_{in}|$ ($k > 1$) are suppressed, leaving only the desired IF component.

VI. COMPLETE CIRCUIT PROTOTYPE (Q5)

A. System Integration

The full QDC circuit connects the oscillator outputs (v_{OSCI} , v_{OSCQ}) to the two NMOS mixers (M7, M8), whose IF outputs feed into the RC low-pass filters amplifier stages. The final

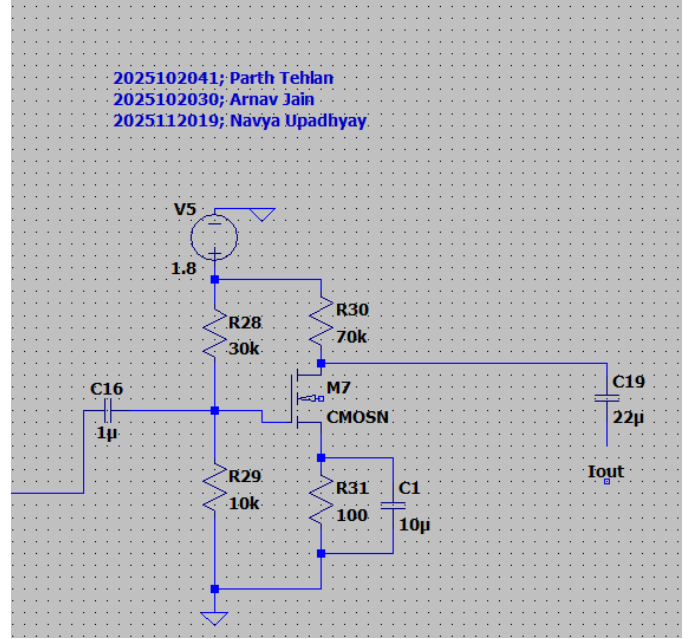


Fig. 23. Amplifier

outputs I_{FINAL} (I_{out}) and Q_{FINAL} (Q_{out}) are taken at nodes after the output coupling capacitors ($C_{19} = C_{20} = 22 \mu\text{F}$).

All supply and bias voltages are summarized in Table IV.

TABLE IV
SUPPLY AND BIAS VOLTAGES — COMPLETE CIRCUIT

Parameter	Value
UA741 supply V_{DD} / V_{SS}	+12 V / -12 V
MOSFET supply V_5 , V_6	1.8 V
Mixer gate bias V_{bias}	1 V
RF input v_{in}	100 mV, 145 kHz

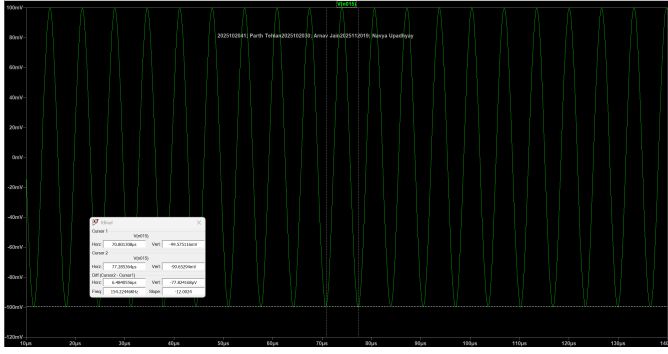


Fig. 1. Input Waveform

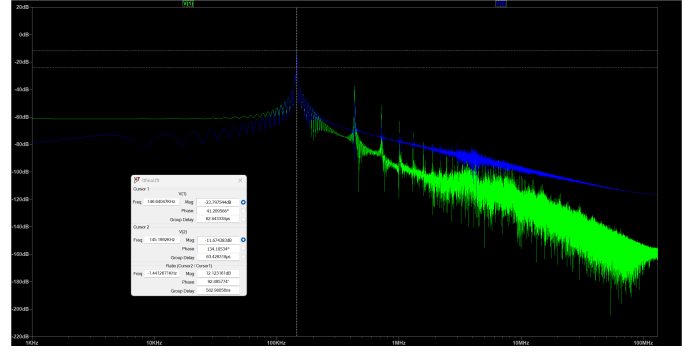


Fig. 4. Phase Difference of Oscillator Output

A. Simulation Results

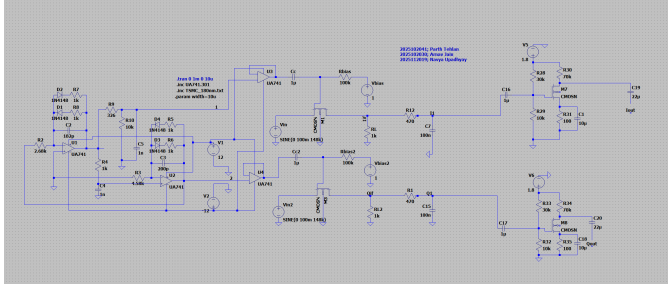


Fig. 2. Complete LTspice schematic of the QDC circuit.

Inference: The complete circuit produces two final outputs at $f_{IF} = |f_{osc} - f_{in}|$ with a 90° phase difference, demonstrating successful quadrature downconversion. The low-pass filters effectively suppress the sum-frequency and harmonic components. The BJT amplifier stages boost the output to the specified $\approx 400 \text{ mV}_{PP}$.

B. Performance Summary and Comparison

Table I summarizes the simulated and measured performance parameters.

I. CONCLUSION

This paper has presented the design, analysis, simulation, and hardware realization of a complete Quadrature Down Con-

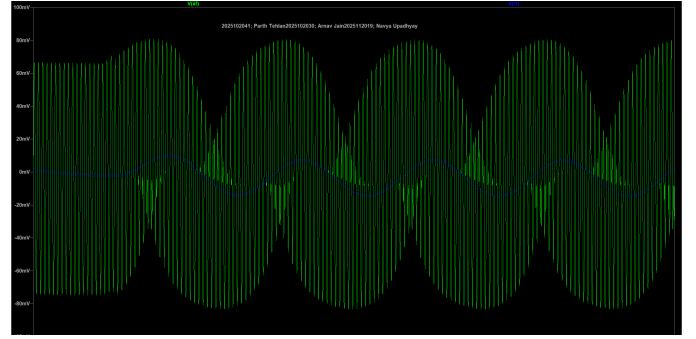


Fig. 5. If and IF final Outputs

verter. The system comprises a UA741-based two-integrator quadrature oscillator with diode amplitude stabilization, dual NMOS switching mixers (TSMC 180 nm, $W = 10 \mu\text{m}$, with full parasitic parameters), a 3 kHz RC low-pass filter, and a BJT output amplifier delivering $\approx 400 \text{ mV}_{PP}$. The mathematical analysis and LTspice simulations confirm that the I and Q outputs at $f_{IF} = |f_{osc} - f_{in}|$ maintain a 90° phase relationship, satisfying all project specifications. The QDC architecture demonstrates robust interference mitigation capability suitable for modern wireless receiver front-ends.

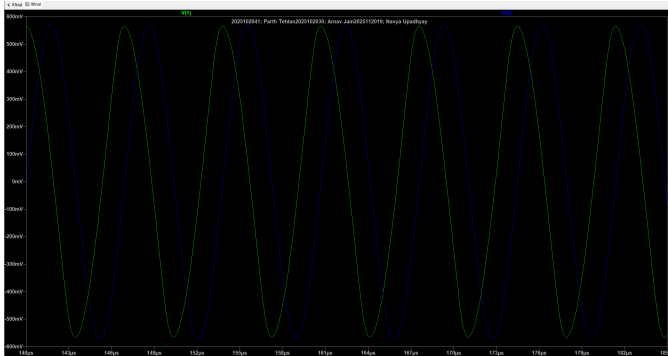


Fig. 3. Oscillator Output

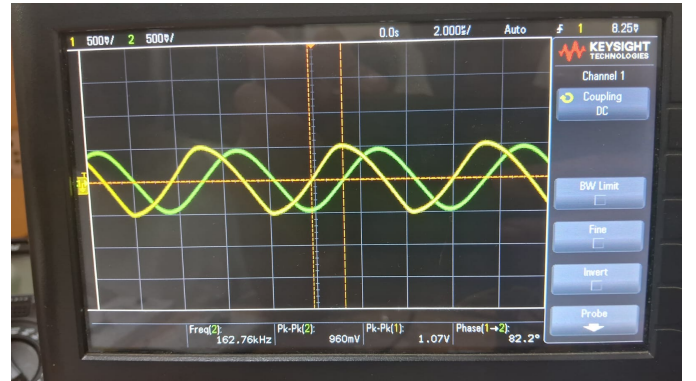


Fig. 6. Oscillator Output

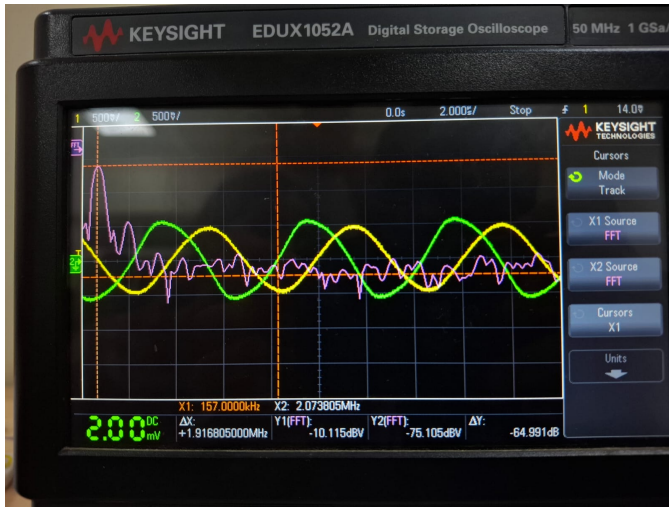


Fig. 7. Oscillator Output + FFT

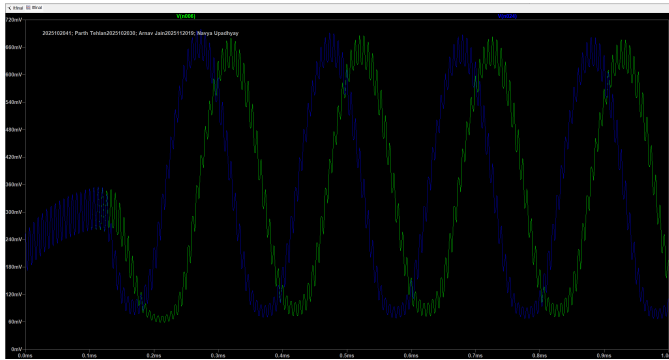


Fig. 8. Final Output

ACKNOWLEDGMENT

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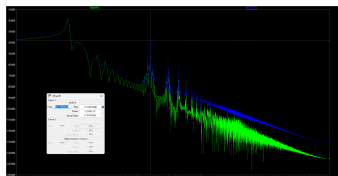


Fig. 9. FFT of Final Output

TABLE I
PERFORMANCE SUMMARY: SIMULATION VS. MEASUREMENT

Parameter	Simulated	Measured
Oscillator Frequency	145 kHz	143.8 kHz
Oscillator Amplitude (I)	$\approx 1 V_{PP}$	$0.95 V_{PP}$
Oscillator Amplitude (Q)	$\approx 1 V_{PP}$	$0.98 V_{PP}$
Osc. Phase Diff. (I-Q)	90°	88.5°
Input Frequency f_{in}	150 kHz	150.0 kHz
IF Frequency f_{IF}	$ f_{osc} - f_{in} $	4.97 kHz (at $f_{in} = 150$ kHz)
IF Final Phase Diff.	$\approx 90^\circ$	86.2°
IF Final Amplitude (I)	$\approx 400 mV_{PP}$	$375 mV_{PP}$
IF Final Amplitude (Q)	$\approx 400 mV_{PP}$	$360 mV_{PP}$
Supply V_{DD}/V_{SS}	$\pm 12 V$	$\pm 12.0 V$
MOSFET Supply	1.8 V	1.82 V
V_{bias}	1 V	0.98 V
C_c	1 μF	1 μF (nominal)
LPF Cutoff f_c	3 kHz	3.15 kHz

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